

Deployment-Oriented Artificial Intelligence

for Battery Management Systems

Neural network ageing estimation, robustness
benchmarking,
embedded optimization, and hardware
integration

vorgelegt von
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Vorwort

This document is a working dissertation mockup. Its purpose is not to provide final prose yet, but to establish a coherent overall narrative, a plausible chapter order, and a first impression of how the dissertation could look when the existing and planned papers are integrated into one thesis manuscript.

Abstract

This dissertation is structured around the deployment-oriented development of artificial intelligence for battery management systems. The planned narrative begins with neural-network-based ageing estimation for stationary storage systems, continues with robustness benchmarking of embedded battery state estimation under disturbed measurements, and then moves toward embedded implementation on microcontrollers using pruning and quantization. A further planned component studies how compression methods interact with different model architectures, while an additional hardware chapter documents the custom battery management system platform and the practical route from algorithm development to embedded execution. The central idea of the dissertation is that battery intelligence should be evaluated not only by nominal prediction accuracy, but by the combined requirement space of estimator performance, hardware feasibility, and implementation effort.

Kurzzusammenfassung

Die Dissertation wird als durchgehende, anwendungsorientierte Geschichte für künstliche Intelligenz im Batteriemanagement aufgebaut. Im Mittelpunkt stehen erstens neuronale Architekturen zur Alterungsbestimmung stationärer Speichersysteme, zweitens die Robustheit eingebetteter Zustandsbestimmung unter realistischen Messstörungen, drittens die effiziente Ausführung solcher Modelle auf Mikrocontrollern durch Pruning und Quantisierung sowie viertens die Einbettung in eine reale Hardware- und BMS-Plattform. Die übergeordnete Leitidee ist, dass KI-gestützte BMS-Methoden nicht nur nach Genauigkeit, sondern entlang der drei Anforderungsfelder Leistung, Hardware und Implementierungsaufwand bewertet werden sollten.

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Abbreviations

AI	Artificial Intelligence
BMS	Battery Management System
ECM	Equivalent Circuit Model
LSTM	Long Short-Term Memory
MCU	Microcontroller Unit
MLP	Multilayer Perceptron
SOC	State of Charge
SOH	State of Health

Symbols

SOC	State of charge
SOH	State of health
Q	Cumulative charge or charge throughput
t	Time
U	Voltage
I	Current

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Introduction and Motivation

Role in the dissertation. This opening chapter should make clear that the dissertation is not a loose bundle of papers. It should define a single scientific story: how battery intelligence moves from data-driven ageing estimation toward robust and resource-efficient embedded deployment inside practical BMS environments.

The overarching motivation of the dissertation is the gap between algorithm-centric battery AI research and deployable BMS intelligence. In many papers, the central question is still whether a model reaches a low error on a clean dataset. In a deployable battery system, however, the more relevant question is whether the method remains useful when it must run on constrained hardware, tolerate imperfect measurements, and fit into a real engineering workflow. This dissertation therefore places prediction quality, robustness, embedded feasibility, and hardware integration into one combined research frame.

1.1 Central Thesis

The central thesis of the dissertation is that AI-enabled BMS functions should be designed and evaluated along three linked requirement dimensions:

- estimator performance, including baseline accuracy, recovery behavior, and robustness against realistic disturbances,
- hardware feasibility, including microcontroller resources, latency, integration constraints, and measurement-chain limitations,
- implementation strategy, including feature engineering, model architecture, compression, and the route from research code to embedded inference.

1.2 Research Questions

1. How can neural-network architectures represent battery ageing in stationary storage systems with sufficient interpretability and practical utility?
2. How do embedded battery state estimators behave when realistic measurement disturbances and timing irregularities are introduced?
3. How can AI-based SOC and SOH estimators be reduced and transferred onto microcontrollers without losing too much predictive utility?
4. How strongly does the effect of quantization and pruning depend on the underlying model architecture?
5. How should a hardware platform and validation chain be structured so that AI-based BMS methods can be tested under realistic embedded conditions?

1.3 Included and Planned Publications

The dissertation integrates the published and planned work listed in Table 1.1. The intended narrative order follows your preferred logic: first the ageing-estimation work, then the robustness benchmark, then the embedded deployment paper, followed by the planned compression study and the hardware-focused integration perspective.[1, 2, 3, 4]

Tabelle 1.1: Mapping of the included and planned work to the dissertation narrative.

Order	Source work	Primary dimension	di- mension	Intended role in the dissertation
1	Neural Network Architecture for Determining the Aging of Stationary Storage Systems in Smart Grids	Model design and ageing estimation		Establishes the data-driven ageing-estimation foundation and motivates the role of feature engineering, temporal context, and model architecture.
2	Performance Benchmarking of Embedded Battery State Estimation	Robustness and deployment-facing performance		Introduces the dissertation-wide requirement space and shows why clean-condition accuracy alone is insufficient.
3	Embedded Artificial Intelligence in Battery Management Systems	Embedded deployment and optimization		Connects model design to actual MCU execution through pruning, quantization, latency, and memory metrics.
4	Planned comparative study on pruning and quantization across architectures	Compression sensitivity and co-design		Extends the embedded perspective from one concrete pipeline toward a broader architecture-dependent design framework.
5	Planned hardware integration chapter around the custom BMS platform	Hardware and validation ecosystem		Anchors the software and model contributions in a real system

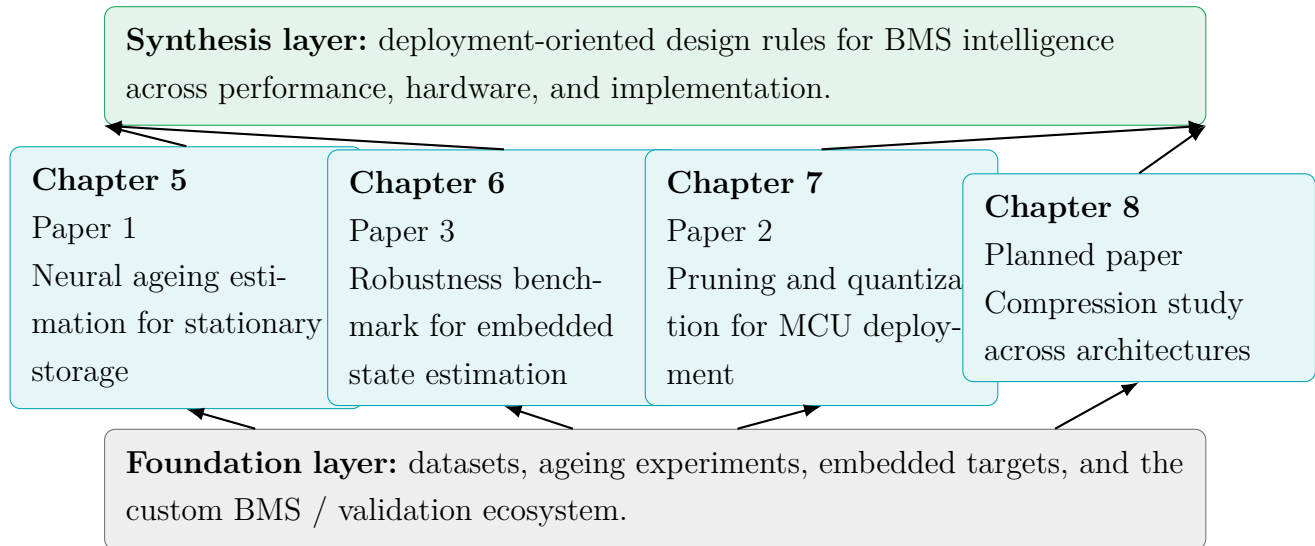


Abbildung 1.1: Proposed dissertation roadmap. The existing papers form the main argumentative chain, while the hardware ecosystem and the planned compression study widen the work from algorithmic performance toward full deployment-oriented BMS intelligence.

Dissertation Framework and Requirement Space

Role in the dissertation. This chapter should introduce the conceptual frame that ties the dissertation together. The requirements graphic from the robustness paper is well suited to become the central figure of the entire thesis.

The strongest unifying concept already available in the current paper set is the requirement-space view from the robustness benchmark. It separates deployment-oriented battery intelligence into performance, hardware, and implementation considerations. This figure is therefore a natural candidate for the dissertation-wide framing chapter, because it prevents the thesis from being read as a simple succession of disconnected paper topics.

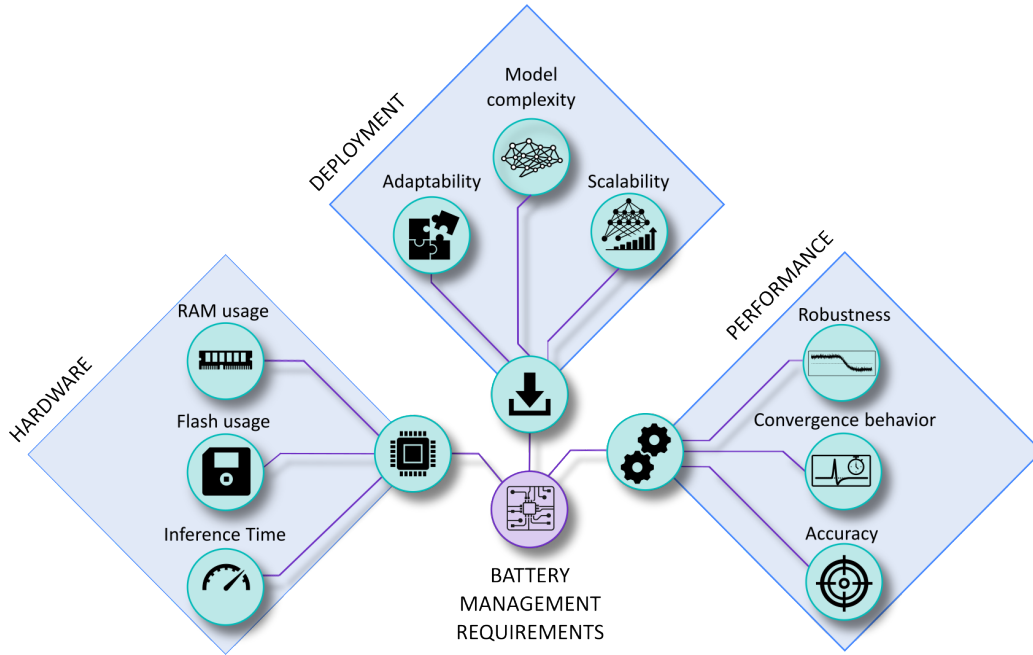


Abbildung 2.1: Working dissertation framing based on the requirement-space figure from the robustness benchmark. In the dissertation, this figure can be reinterpreted as the global structure that connects algorithm design, embedded feasibility, and hardware integration.

2.1 Performance Dimension

This dimension captures nominal estimator quality, robustness under disturbed measurements, and synchronization behavior after mismatched initial conditions or missing data. It is primarily represented by the robustness benchmark chapter, but it should also be used as a lens for interpreting the embedded and ageing-estimation chapters.

2.2 Hardware Dimension

This dimension captures what the algorithm must respect once it is placed into a BMS setting: sensing, memory, computation time, microcontroller capabilities, and the practical route toward a testable prototype. This dimension justifies the later chapters on MCU execution and the custom BMS platform.

2.3 Implementation Dimension

This dimension captures how the method is actually built: architecture choice, temporal context, feature engineering, compression strategy, and software-to-firmware transfer. It links the SOH architecture work, the embedded AI work, and the planned cross-architecture compression study.

2.4 How the Thesis Should Read as One Story

The dissertation should not be organized around journals or submission histories. Instead, it should read from left to right through the requirement space:

- first, how a neural model can capture ageing information in the battery domain,
- second, why estimator quality must be judged under realistic disturbances,
- third, how the resulting models can be translated into efficient embedded inference,
- fourth, how a real hardware ecosystem can host, validate, and extend such methods.

State of the Art and Technical Background

Role in the dissertation. This chapter should provide the common vocabulary for all later chapters and prevent repeated background sections inside the integrated paper material.

3.1 Battery Management Systems

This section should introduce battery-system functions, the role of the BMS, and the relation between sensing, control, safety, and diagnosis. It should define how SOC and SOH support energy management, operating limits, charging strategies, and lifetime-aware operation.

3.2 Neural Networks for Battery Ageing and State Estimation

This section should summarize why neural methods are attractive in battery applications: nonlinear representation power, temporal modeling, data-driven feature fusion, and the flexibility to capture ageing-related patterns that are difficult to parameterize analytically. It should also acknowledge the trade-off between accuracy, interpretability, and deployment effort.

3.3 Embedded AI in Battery Applications

This section should move from model quality toward deployment quality. The focus is not only on prediction error but also on Flash, RAM, latency, reproducibility,

numerical stability, and the workflow needed to move from Python-based experimentation to resource-constrained MCU inference.

3.4 Compression and Model-Hardware Co-Design

This section should prepare the ground for the embedded paper and the planned future paper. It should explain why pruning and quantization are not merely afterthought optimizations, but central design choices once AI is treated as a practical BMS component.

Experimental Basis and Hardware Ecosystem

Role in the dissertation. This chapter provides the practical basis shared by the later thesis chapters. It should become the place where the datasets, the experimental environment, the embedded execution targets, and the custom BMS platform are described once and then referenced throughout the dissertation.

4.1 Data Sources and Ageing Campaigns

This section should combine the ageing data story behind the stationary-storage SOH work with the datasets used for the embedded SOC/SOH and robustness studies. The goal is not to force all experiments into one dataset narrative, but to make transparent which data sources support which research question.

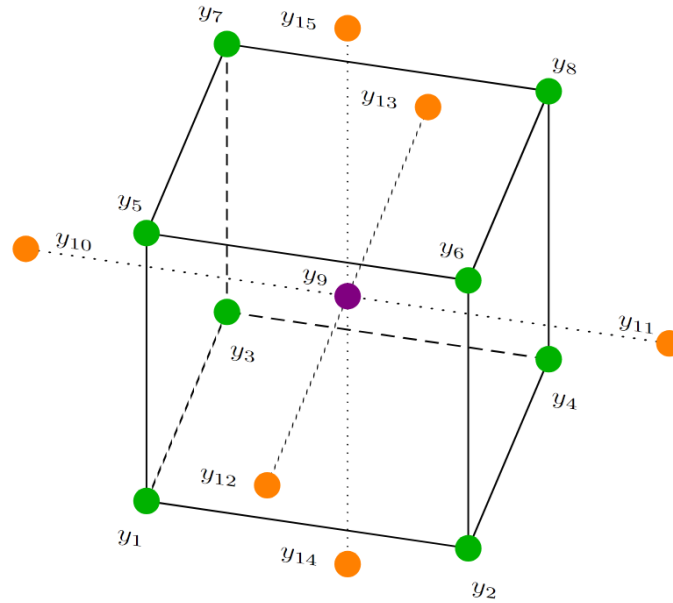


Abbildung 4.1: Existing design-of-experiments visualization that can be reused in the dissertation to anchor the data and ageing-test foundation.

4.2 Embedded Targets and Execution Chain

This section should explain where the embedded execution results come from: the development environment, the selected microcontrollers, the software stack, and the measurement strategy for latency and memory. It should form the bridge between the robustness benchmark and the embedded AI chapter.

4.3 Custom BMS Platform

The custom BMS platform deserves a dedicated subsection or even a later stand-alone expansion, because it turns the dissertation from an algorithm study into a system study. At this stage, the structure should reserve enough space for:

- the board architecture and sensing concept,
- communication interfaces and measurement chain,
- the role of the platform in validating AI-based state estimation,
- the practical difference between the self-built BMS and the MCU dev-board experiments.

Planned figure placeholder

Photograph or block diagram of the self-built BMS platform, sensor front-end, communication interfaces, host tooling, and the relation to the MCU dev-board execution environment. This is the right place to later document why the hardware chapter matters even if some state-estimation runs were executed on a separate development board.

Abbildung 4.2: Reserved figure placement for the hardware chapter and the custom BMS ecosystem.

4.4 Recommended Writing Logic

To avoid repetition later, this chapter should be written as shared infrastructure:

- data and experiments first,
- then embedded execution targets,
- then the custom BMS platform,
- then a short summary of which later chapter depends on which part of this ecosystem.

Neural Network Architecture for Ageing Estimation in Stationary Storage Systems

Role in the dissertation. This chapter should integrate the first paper into the dissertation as the opening methodological contribution. It establishes the data-driven ageing-estimation viewpoint and introduces the neural-architecture perspective that later reappears in the embedded chapters.

5.1 Core Contribution

The core contribution of the first paper is the architecture-driven estimation of battery ageing in stationary storage systems. Within the dissertation, this chapter should be presented not only as a stand-alone SOH estimation study, but also as the first demonstration of how architecture choice, temporal context, and feature engineering affect deployment-relevant battery intelligence.[1]

5.2 Material to Reuse from the Paper

- dataset description and scenario design for stationary storage ageing,
- preprocessing and feature-engineering pipeline,
- lag-sequence construction and temporal representation,
- MLP architecture and training setup,
- result interpretation for typical and atypical ageing trajectories.

5.3 Dissertation-Specific Expansion

This chapter can be strengthened beyond the original paper by explicitly connecting it to the later BMS requirements view:

- Why is ageing estimation relevant beyond pure SOH prediction?
- Which parts of the pipeline are transferable to embedded BMS state estimation?
- Which parts are data-rich and stationary-storage-specific, and which parts generalize?
- How does this chapter motivate later questions about robustness and deployment?

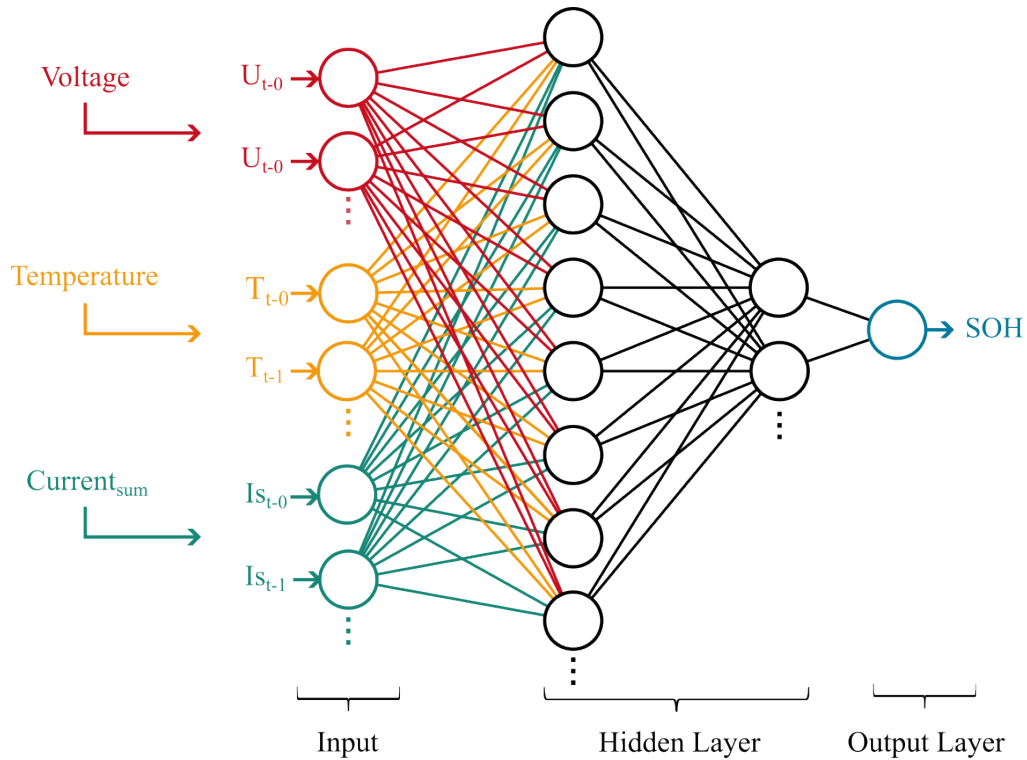


Abbildung 5.1: Working architecture figure for the chapter derived from the first paper. In the dissertation, this figure can anchor the discussion of representation choice, temporal context, and feature engineering for ageing estimation.

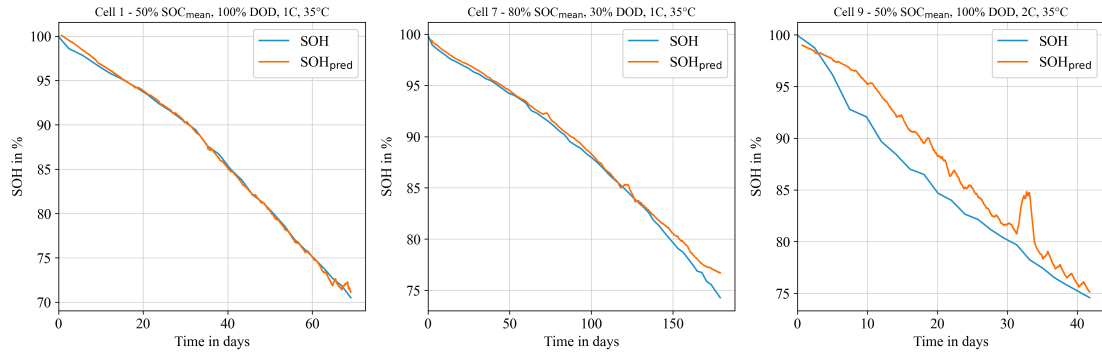


Abbildung 5.2: Representative result figure from the first paper. This can be used to show that the dissertation starts from a strong data-driven ageing-estimation perspective before moving toward robustness and embedded deployment.

Robustness Benchmarking of Embedded Battery State Estimation

Role in the dissertation. This chapter should act as the conceptual pivot of the dissertation. It moves the story from pure model quality toward deployment-facing performance and provides the requirement-space figure that can structure the thesis as a whole.

6.1 Core Contribution

The core contribution of the robustness benchmark is the controlled comparison of different embedded-ready estimator classes under measurement disturbances, missing samples, and initialization mismatch. In the dissertation, this chapter should be framed as the point where the evaluation logic changes from nominal model accuracy toward real deployment conditions.[2]

6.2 Material to Reuse from the Paper

- the requirement-space framing,
- the disturbance taxonomy and benchmark methodology,
- the matched comparison between direct-measurement, hybrid, and data-driven estimators,
- the decision-oriented interpretation of robustness results,

- the conclusion that no single estimator dominates all deployment-relevant dimensions.

6.3 Dissertation-Specific Expansion

Inside the dissertation, this chapter should do more than reproduce the paper:

- explicitly connect robustness to later embedded implementation constraints,
- connect disturbance sensitivity to hardware and sensor-chain realities,
- interpret robustness as a selection criterion for future compression and deployment decisions,
- use the findings as a bridge from algorithm benchmarking to hardware-aware design.

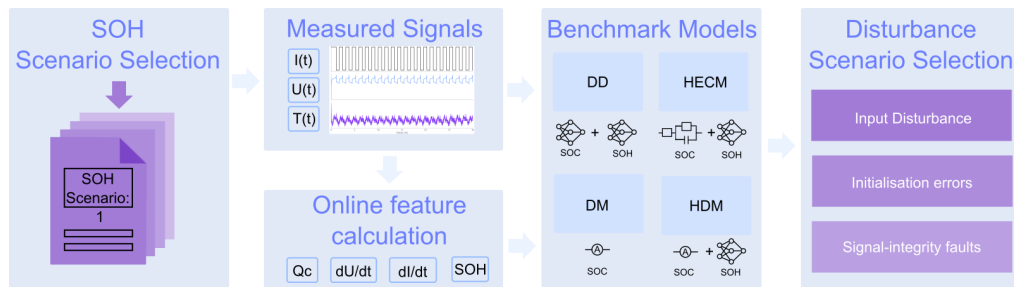


Abbildung 6.1: Methodology overview from the robustness benchmark. In the dissertation, this figure should become the anchor for the deployment-facing evaluation philosophy.

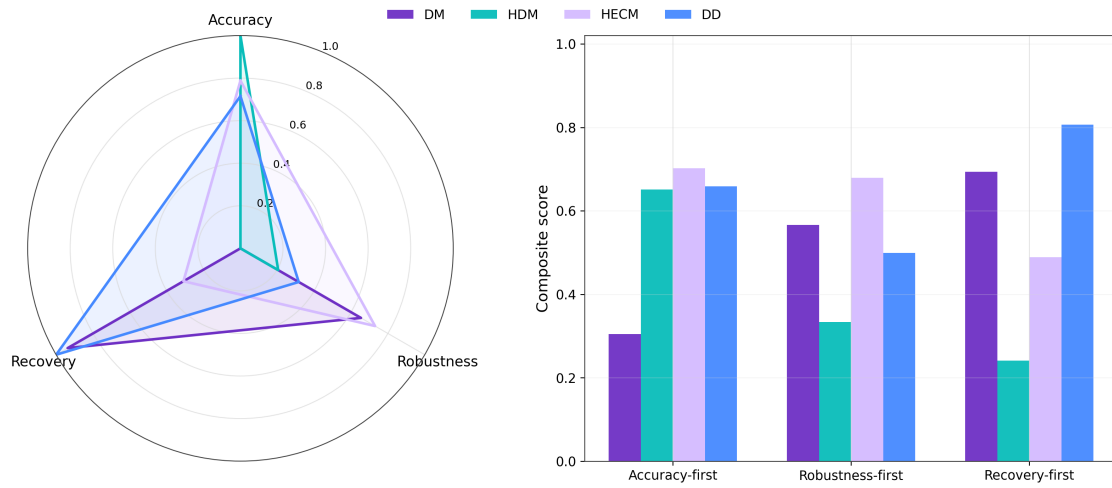


Abbildung 6.2: Decision-oriented synthesis from the robustness benchmark. This fits naturally into the dissertation because it already translates experimental results into estimator-selection guidance.

Embedded Artificial Intelligence for SOC and SOH Estimation on Microcontrollers

Role in the dissertation. This chapter should integrate the embedded paper as the bridge from battery AI models to resource-constrained execution. It is the chapter in which the dissertation becomes explicitly about embedded implementation quality, not only about estimator behavior.

7.1 Core Contribution

The embedded AI paper contributes the deployment path from floating-point sequence models toward practical MCU inference through pruning, quantization, and on-device KPI analysis. Within the dissertation, this chapter should establish that architecture design and embedded execution cannot be separated once the target application is a BMS.[3]

7.2 Material to Reuse from the Paper

- the SOC/SOH model pipeline and streaming inference setup,
- the pruning and quantization methodology,
- the STM32-based measurement workflow,
- Flash, RAM, and latency KPIs,
- the trade-off analysis between resource savings and predictive quality.

7.3 Dissertation-Specific Expansion

To integrate smoothly into the dissertation, this chapter should emphasize:

- how the embedded results depend on the robustness perspective introduced earlier,
- how compression is constrained by estimator architecture and task structure,
- how on-device execution metrics should influence model selection,
- how the dev-board deployment relates to the broader BMS hardware ecosystem.

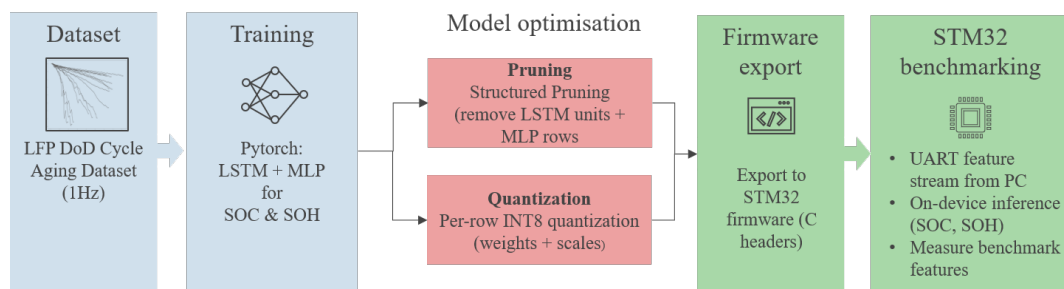


Abbildung 7.1: Embedded deployment pipeline reused as a dissertation anchor figure. This figure can connect software training, model export, and MCU execution in one visual narrative.

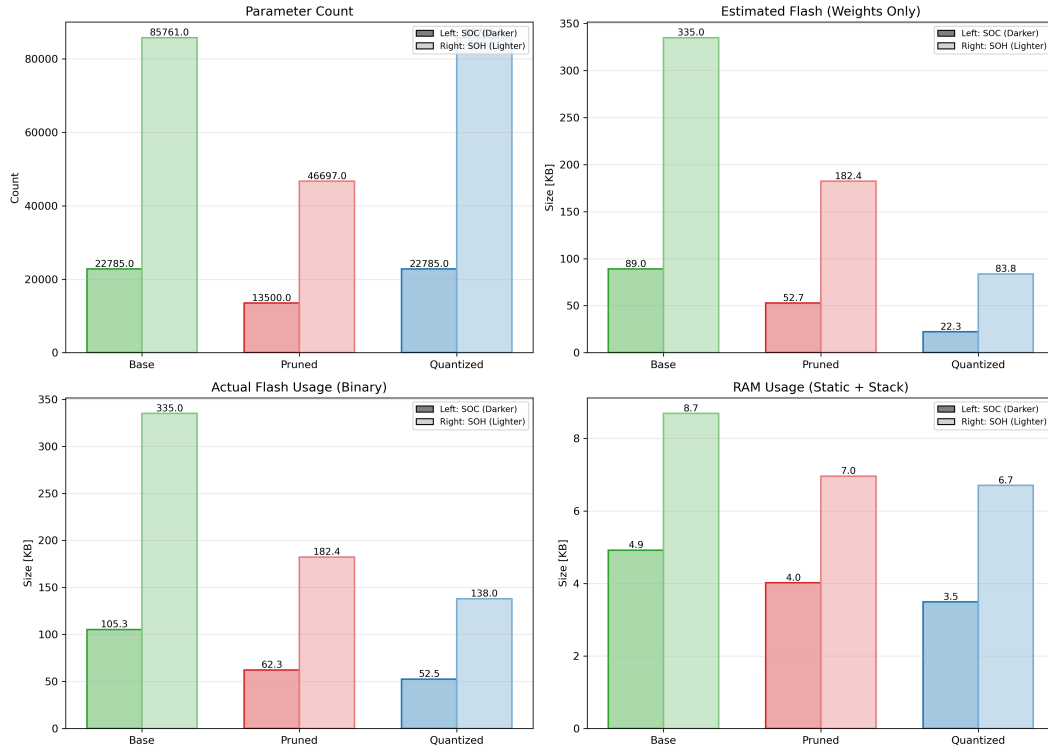


Abbildung 7.2: Resource landscape for the embedded AI chapter. In the dissertation, this type of figure will help position resource efficiency as a first-class evaluation dimension rather than an appendix metric.

Planned Comparative Study on Pruning and Quantization Across Model Architectures

Role in the dissertation. This planned chapter is the cleanest place to introduce the next paper you described. It should extend the current embedded results from one concrete pipeline toward a broader model-compression design framework.

8.1 Why This Chapter Matters

The existing embedded paper already shows that pruning and quantization can be useful. The next logical step is to ask whether those gains hold equally across very different model families. This is where the dissertation can move beyond a single embedded case study and develop genuinely transferable design rules.

8.2 Recommended Scope

The planned comparison chapter should explicitly vary both architecture and compression strategy. A useful comparison matrix would likely include:

- feed-forward or lightweight MLP baselines,
- recurrent models such as GRU or LSTM,
- hybrid models with explicit state or physics-based structure,
- multiple pruning ratios and multiple quantization schemes,

- shared embedded KPIs and task-specific accuracy metrics.

Tabelle 8.1: Proposed experiment matrix for the planned compression chapter.

Model family	Main task		Compression variables	Expected dissertation value
MLP	SOH or static surrogate tasks		INT8, mixed precision, structured pruning	Simple low-cost baseline and reference point for interpretability and resource use.
GRU / LSTM	Streaming SOC or SOH		hidden-size pruning, INT8 export, activation range analysis	Direct continuation of the current embedded AI work.
Hybrid neural / physics-aware models	SOC correction or joint estimation		selective compression of neural sub-blocks	Useful for understanding whether hybrid structures compress differently from purely neural ones.
Tiny deployment baselines	MCU-only	inference	aggressive compression and latency-first tuning	Practical lower-bound reference for real BMS integration.

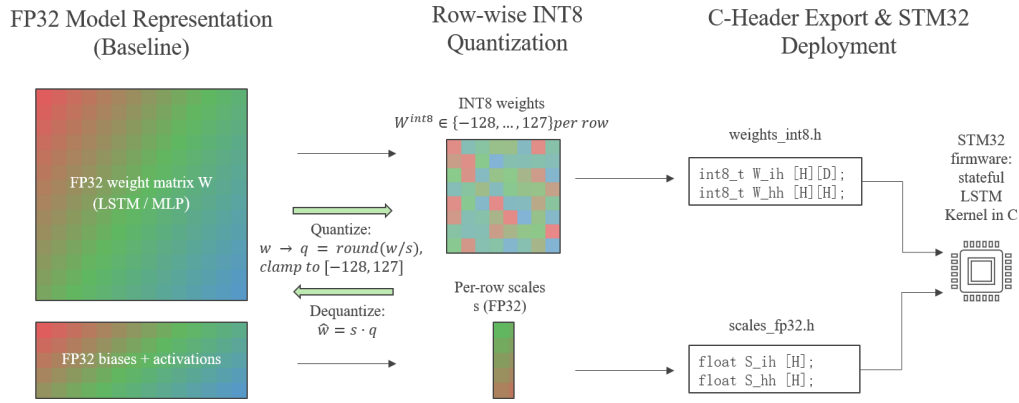


Abbildung 8.1: Existing quantization schematic reused as the visual starting point for the planned architecture-comparison chapter.

8.3 How This Chapter Fits the Overall Thesis

This future chapter should not be presented as an isolated extra paper. It is the natural continuation of the embedded story:

- Chapter 5 asks how architectures represent ageing information.
- Chapter 6 asks how estimator classes behave under disturbance.
- Chapter 7 asks how one embedded pipeline behaves under compression.
- Chapter 8 generalizes the compression question across architectures.

Cross-Paper Synthesis and Design Guidelines

Role in the dissertation. This synthesis chapter should make the dissertation more valuable than the sum of the included papers. It is the place where the three requirement dimensions are turned into practical design rules.

9.1 Integrated Design Perspective

The dissertation should culminate in a design-oriented synthesis chapter rather than stop at paper-specific summaries. This chapter can explicitly state what the combined work teaches about choosing battery-AI methods under practical deployment constraints.

9.2 Proposed Synthesis Matrix

Tabelle 9.1: Proposed synthesis matrix that links dissertation chapters to the three central requirement dimensions.

Dissertation contribution	contri-	Performance	Hardware	/	Implementation logic
Neural estimation chapter	ageing-	high relevance for representation quality	indirect relevance	rele-	strong relevance for architecture and feature design
Robustness benchmark chapter	bench-	strongest relevance	medium relevance through sensing realism	rele-	strong relevance for evaluation methodology
Embedded AI chapter	AI chap-	medium relevance	strongest relevance	rele-	strongest relevance for compression and deployment workflow
Planned compression chapter	compression	medium relevance	strongest relevance	rele-	strongest relevance for model-hardware co-design
Hardware ecosystem chapter	ecosystem	indirect relevance	strongest relevance	rele-	medium relevance through system integration

9.3 Candidate Final Design Rules

This final synthesis chapter is the right location for concise design rules such as:

- do not choose estimators by clean-condition accuracy alone,
- evaluate robustness before optimizing for compression,
- treat quantization and pruning as architecture-dependent design choices,
- separate dataset quality, estimator quality, and embedded execution quality,
- connect all algorithmic claims to a credible hardware and validation chain.

9.4 Why the Dissertation Benefits from This Structure

This organization allows each paper to keep its own technical identity while avoiding three common cumulative-dissertation problems:

- repeated background across chapters,
- unclear progression from one publication to the next,
- missing synthesis beyond already published paper conclusions.

Role in the dissertation. The concluding chapter should close the dissertation at the synthesis level rather than only repeat chapter summaries.

The final dissertation should conclude that deployment-oriented battery AI emerges only when model design, robustness, embedded execution, and hardware integration are treated as one linked problem. The current structure already supports that conclusion by giving each paper a clear role inside a broader argument.

10.1 Immediate Next Writing Steps

- decide whether the final dissertation language will remain fully English,
- finalize the chapter order and the chapter titles,
- decide how much of each paper is reused almost verbatim and how much is rewritten,
- collect the available hardware material for the custom BMS chapter,
- define the concrete experiment matrix for the planned compression paper.

10.2 Likely Final Contribution Statement

The likely final contribution of the dissertation is a deployment-oriented framework for AI-enabled battery management systems that spans:

- neural ageing estimation,
- robustness-oriented estimator evaluation,

- resource-constrained embedded inference,
- model-compression co-design,
- and practical hardware integration.

Literatur

- [1] Florian Rzepka, Philipp Hematty, Mano Schmitz und Julia Kowal. *Neural Network Architecture for Determining the Aging of Stationary Storage Systems in Smart Grids*. First dissertation paper. 2023.
- [2] Florian Rzepka und Julia Kowal. *Performance Benchmarking of Embedded Battery State Estimation: A Measurement-Only Comparison of Direct-Measurement, Hybrid, and Data-Driven Estimators*. Robustness benchmark manuscript. 2026.
- [3] Florian Rzepka und Julia Kowal. *Embedded Artificial Intelligence in Battery Management Systems: Pruning and Quantization for Efficient State-of-Charge and State-of-Health Estimation*. Embedded AI manuscript. 2025.
- [4] Florian Rzepka und Julia Kowal. *Planned comparative study on pruning and quantization across embedded BMS model architectures*. Planned follow-up paper. 2026.

Reserved Appendix Space

A

This appendix is intentionally left as a placeholder for supplementary tables, additional hardware documentation, experiment matrices, and publication-specific material that should not interrupt the flow of the main dissertation.